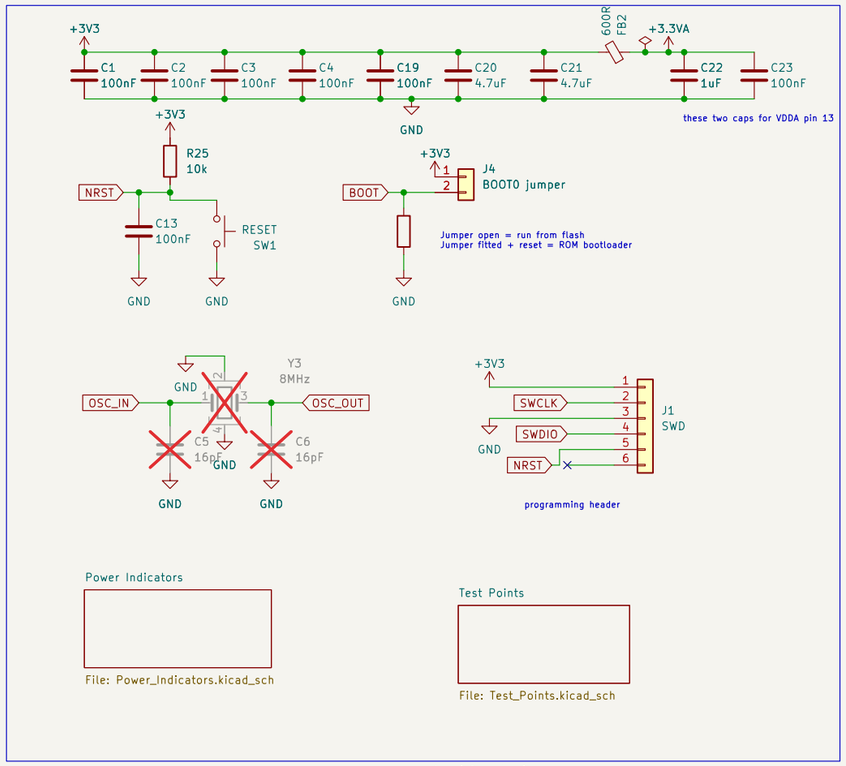
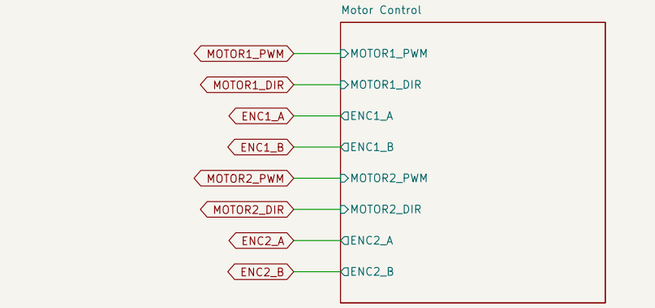
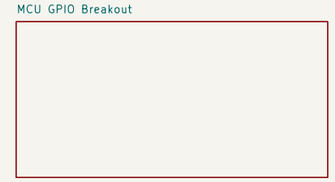
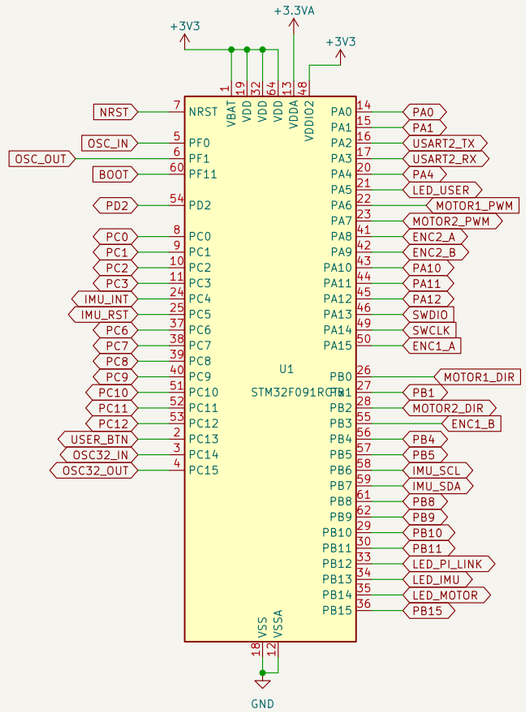
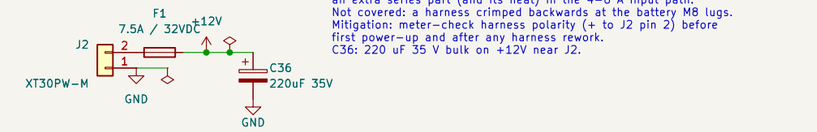




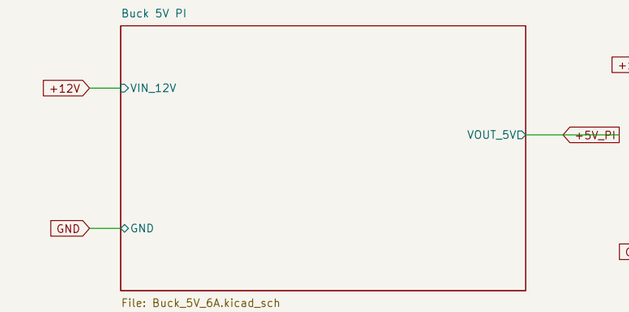


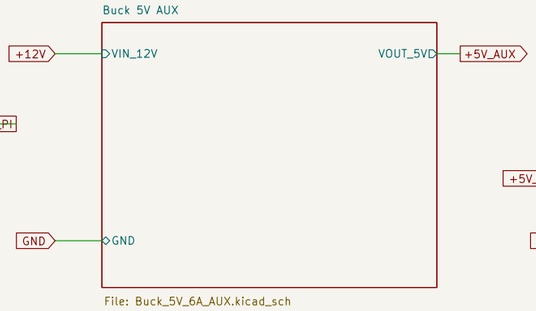


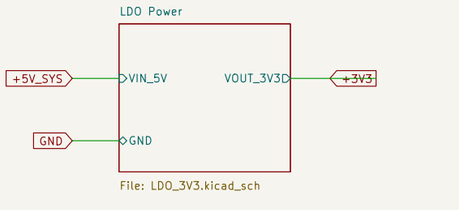


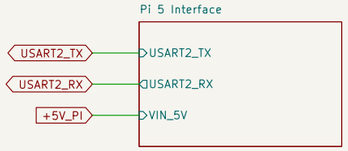


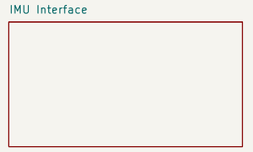












ECE49022 Team 22 senior design – STM32F091 control, power and I/O
Derived from ST NUCLEO-64 MB1136 rev C-04, sheets 2–3

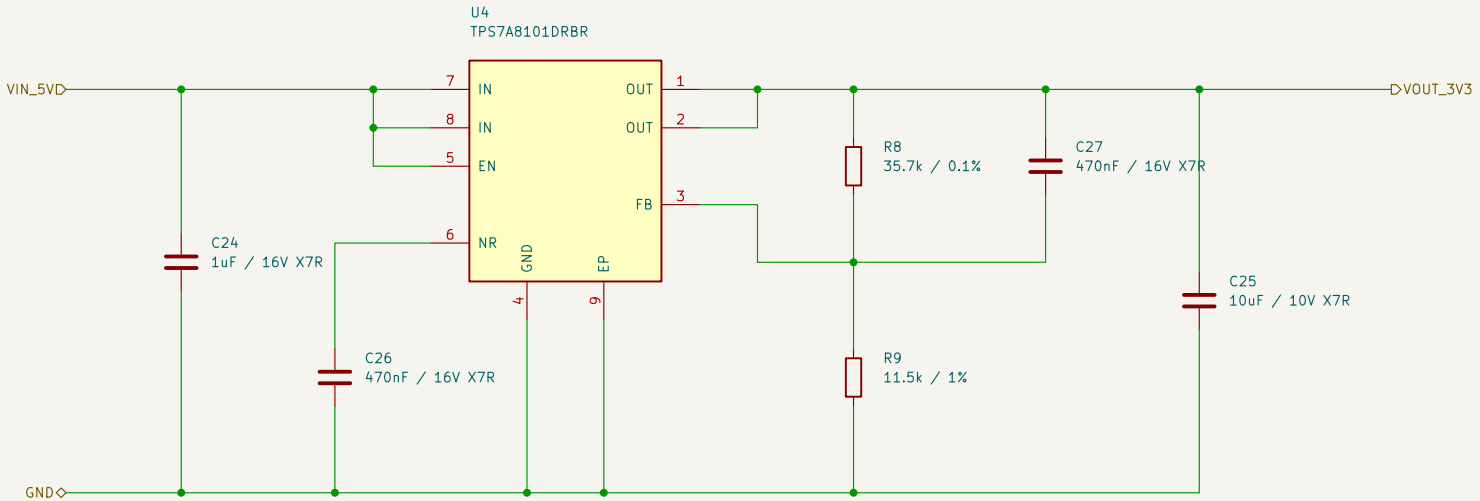
Sheet: /
File: delivery_robot_1.kicad_sch

Title: Delivery Robot Main Board

Size: A3	Date: 2026-10-08	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/11

5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND



VOUT nominal = 0.8 x (1 + 35.7k / 11.5k) = 3.284 V (E96 values)
1 A is the TOTAL output budget for STM32 + IMU + other 3.3 V loads.

C2: require >= 4.7uF EFFECTIVE capacitance; selected part modeled at 8.8uF.
EN tied to IN. C3 sets noise reduction / soft start; C4 bypasses R1.

Layout: capacitors close to U1; EP pad 9 to GND copper and thermal vias.
Heat at 1 A: about 1.7 W at 5 V input; verify junction temperature on final PCB.

5 V motor / encoder supply branches BEFORE this sheet.
Adafruit BNO085 breakout: 3.3 V rail to VIN; 3Vo is an output, leave separate.
Bare BNO085: connect 3.3 V to VDD and VDDIO with its specified decoupling.

5 V to 3.3 V, 1 A total rail capacity

Senior Design

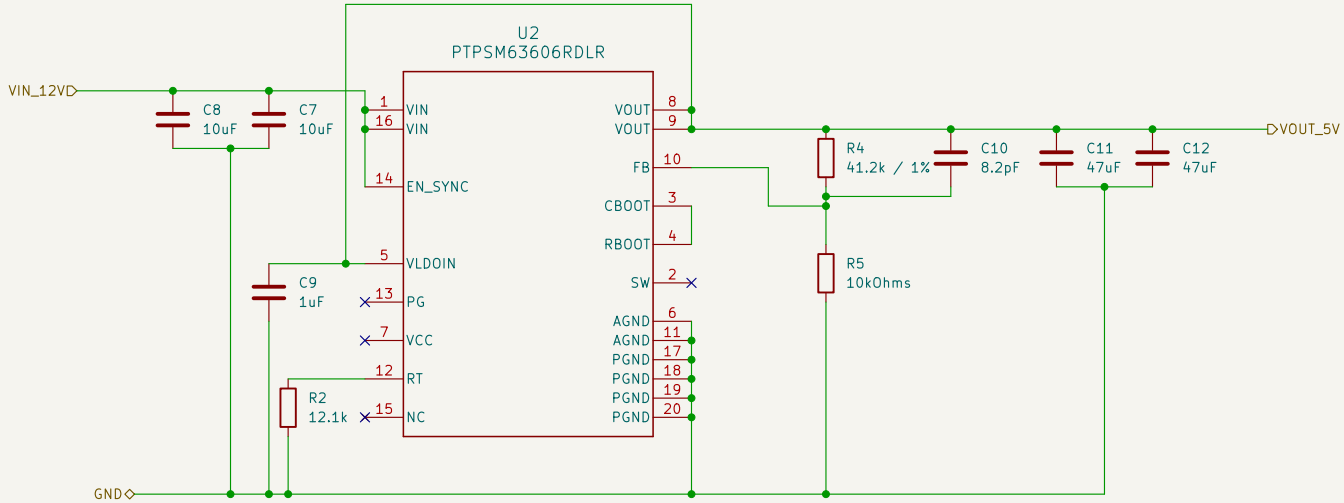
Sheet: /LDO Power/
File: LDO_3V3.kicad_sch

Title: LDO power supply – reusable sheet

Size: A4	Date: 2026-10-05	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/11

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U2)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R4/R5 divider on FB; RT = R2; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U2), 6 A rail

Senior Design

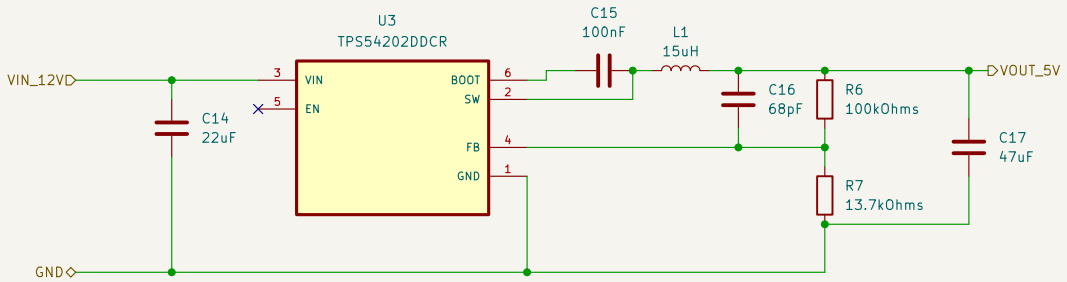
Sheet: /Buck 5V PI/
File: Buck_5V_6A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4	Date: 2026-10-06	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 3/11

12 V – 5 V, 2A Buck Convertor

12 V to 5 V buck converter – TPS54202 (U3)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND

12 V to 5 V buck, TPS54202 (U3), 2 A rail

Senior Design

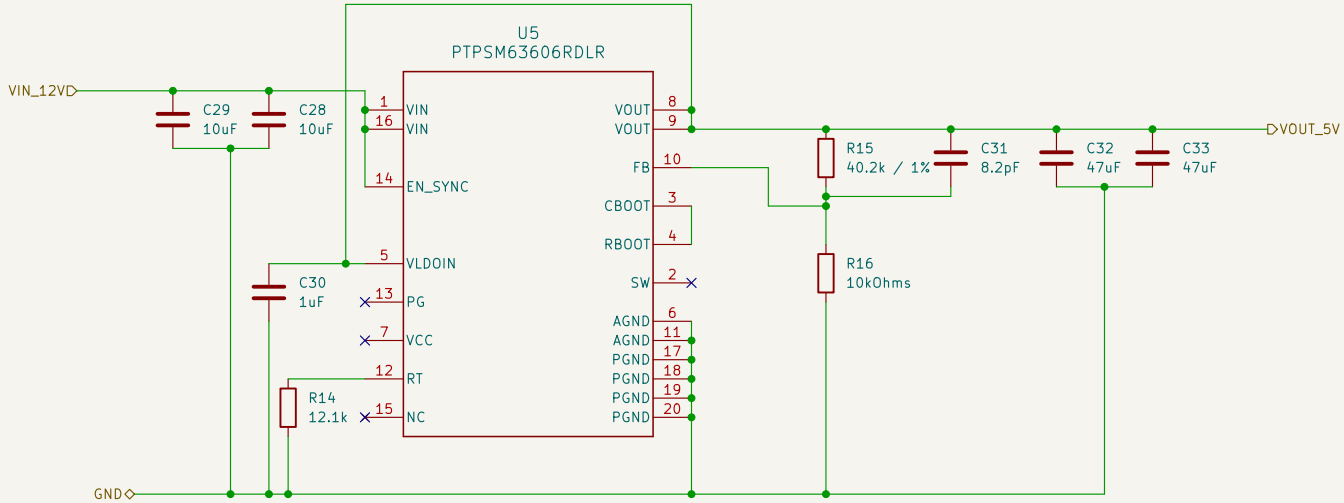
Sheet: /Buck 5V SYS/
File: Buck_5V_2A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4	Date: 2026-10-06	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/11

12 V – 5 V, 6A Buck Convertor

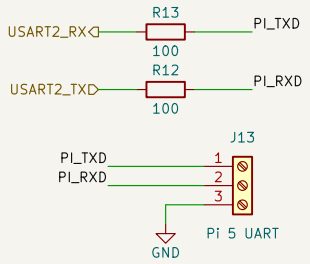
12 V to 5 V buck converter – TPSM63606 (U5)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R15/R16 divider on FB; RT = R14; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

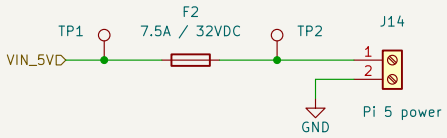
12 V to 5 V buck, TPSM63606 (U5), 6 A rail
Senior Design
Sheet: /Buck 5V AUX/
File: Buck_5V_6A_AUX.kicad_sch
Title: Buck converter – reusable sheet
Size: A4 | Date: 2026-10-06 | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 5/11

Raspberry Pi 5 UART (USART2, PA2 TX / PA3 RX)



J13 1 <- Pi GPIO14 TXD (header pin 8)
J13 2 -> Pi GPIO15 RXD (header pin 10)
J13 3 = GND (Pi header pin 6); 3.3 V logic
R12/R13 near U1 (no ESD array fitted)

Pi 5 power: +5V_PI -> F2 isolation fuse -> J14 USB-C pigtail



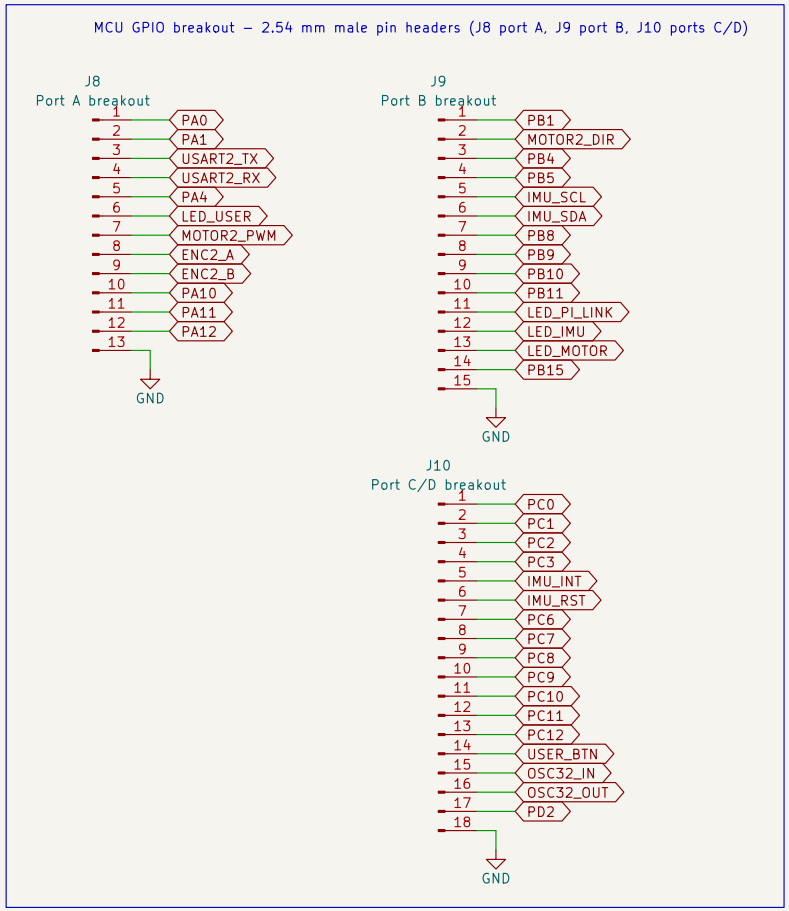
Bring-up: F2 out, check TP1; fit F2, load-test J14; then connect the Pi. 5-6 A path

Senior Design

Sheet: /Pi 5 Interface/
File: Pi5_Interface.kicad_sch

Title: Raspberry Pi 5 interface — UART and power

Size: A4	Date: 2026-10-07	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 6/11



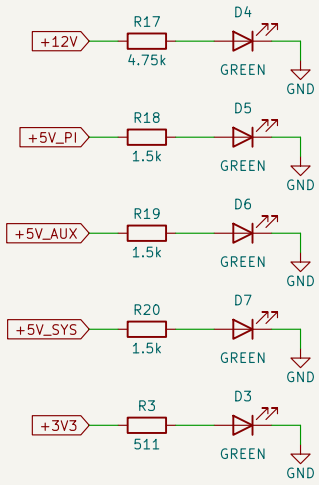
Senior Design

Sheet: /MCU GPIO Breakout/
File: GPIO_Breakout.kicad_sch

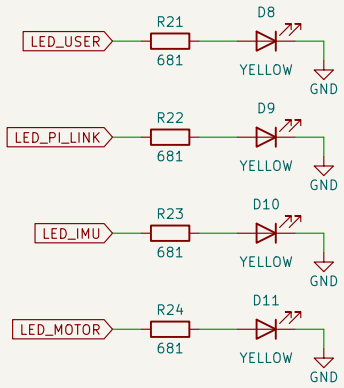
Title: STM32 GPIO breakout headers

Size: A4	Date: 2026-10-07	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 7/11

Power rail indicators — one green LED per rail
(Lite-On LTST-C190KGKT, 0603, Vf 2.0 V typ)



Debug LEDs — STM32 GPIO, active high (Lite-On LTST-C190KSKT yellow, 0603)



PA5 LED_USER: heartbeat
PB12 LED_PI_LINK: Pi UART activity
PB13 LED_IMU: IMU data-ready
PB14 LED_MOTOR: motor running / fault
681R: (3.3 − 2.1) / 681 = 1.8 mA per pin

LED current ~2–2.7 mA: R = (Vrail − 2.0 V) / I. +12V uses 4.75k for the 10.8–14.6 V battery range.
All resistors 0805; worst-case dissipation 34 mW (R17 at 14.6 V).

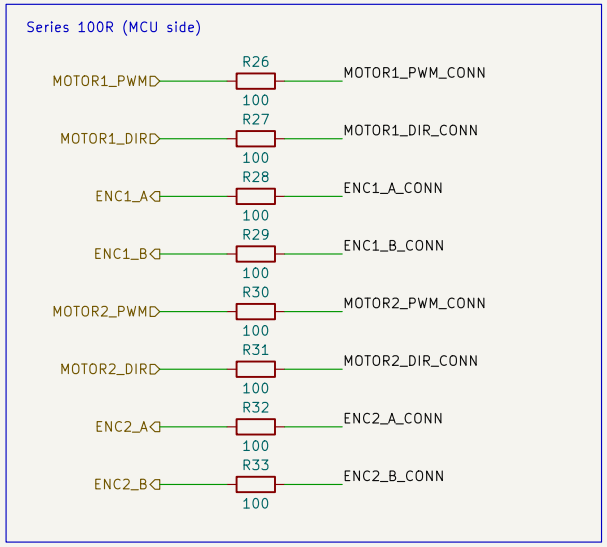
Senior Design

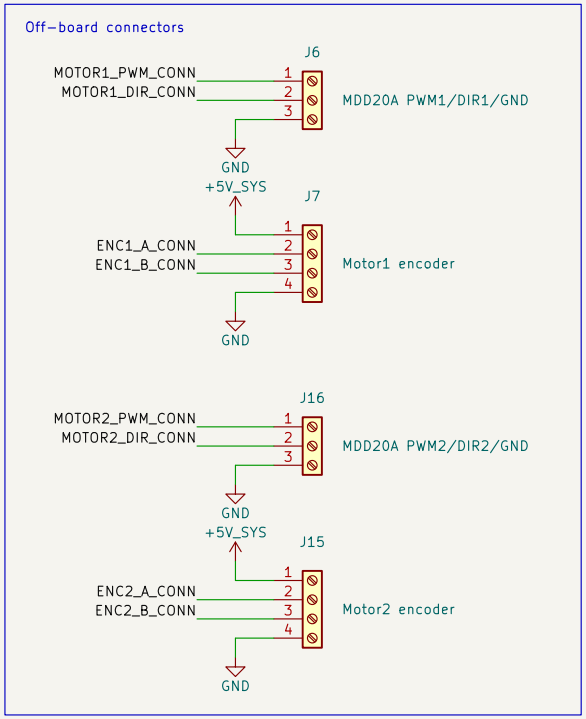
Sheet: /Power Indicators/
File: Power_Indicators.kicad_sch

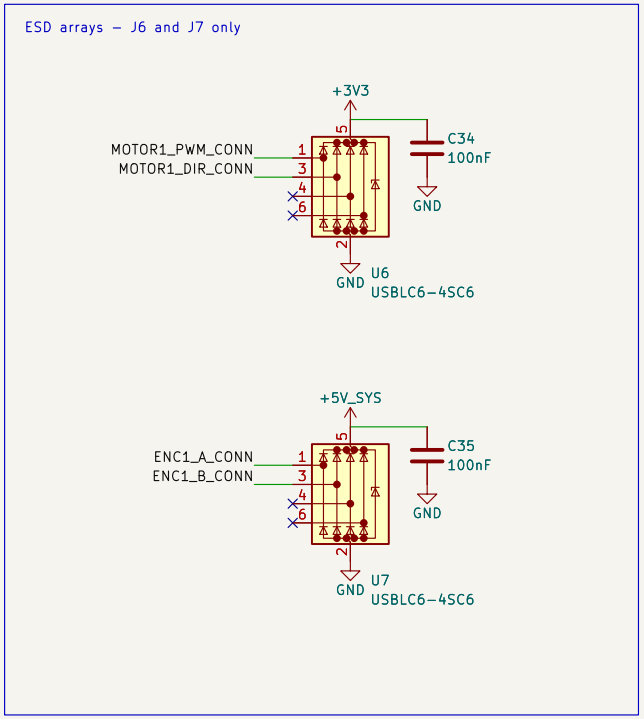
Title: Power rail and debug indicator LEDs

Size: A4	Date: 2026-10-07	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 8/11

Motor control — Cytron MDD20A dual-channel driver + two 37D encoder motors



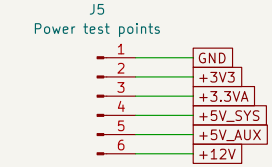





```
STM32F091 PIN ASSIGNMENT          series R    connector
  MOTOR1_PWM  PA6   TIM3_CH1    R26 100R    J6 pin 1 -> MDD20A PWM1
  MOTOR1_DIR  PB0   GPIO        R27 100R    J6 pin 2 -> MDD20A DIR1
  ENC1_A      PA15  TIM2_CH1    R28 100R    J7 pin 2 <- encoder A (yellow)
  ENC1_B      PB3   TIM2_CH2    R29 100R    J7 pin 3 <- encoder B (white)
  MOTOR2_PWM  PA7   TIM3_CH2    R30 100R    J16 pin 1 -> MDD20A PWM2
  MOTOR2_DIR  PB2   GPIO        R31 100R    J16 pin 2 -> MDD20A DIR2
  ENC2_A      PA8   TIM1_CH1    R32 100R    J15 pin 2 <- encoder A (yellow)
  ENC2_B      PA9   TIM1_CH2    R33 100R    J15 pin 3 <- encoder B (white)
Timers: TIM3 = both PWMs (<= 20 kHz for MDD20A); TIM2 = encoder 1, TIM1 = encoder 2.
ESD: USBLC6-4SC6 arrays on motor 1 only — U6 on J6 PWM/DIR (VBUS +3V3), U7 on J7 encoder A/B (VBUS +5V_SYS). Encoders powered from +5V_SYS;
A/B pins must be 5 V tolerant (FT). Battery and motor leads wire directly to the MDD20A.
```

Senior Design		
Sheet: /Motor Control/		
File: Motor_Control.kicad_sch		
Title: Motor control — MDD20A driver and encoders		
Size: A4	Date: 2026-10-08	**Rev: 1.1**
KiCad E.D.A. 10.0.6		Id: 9/11

Power test points

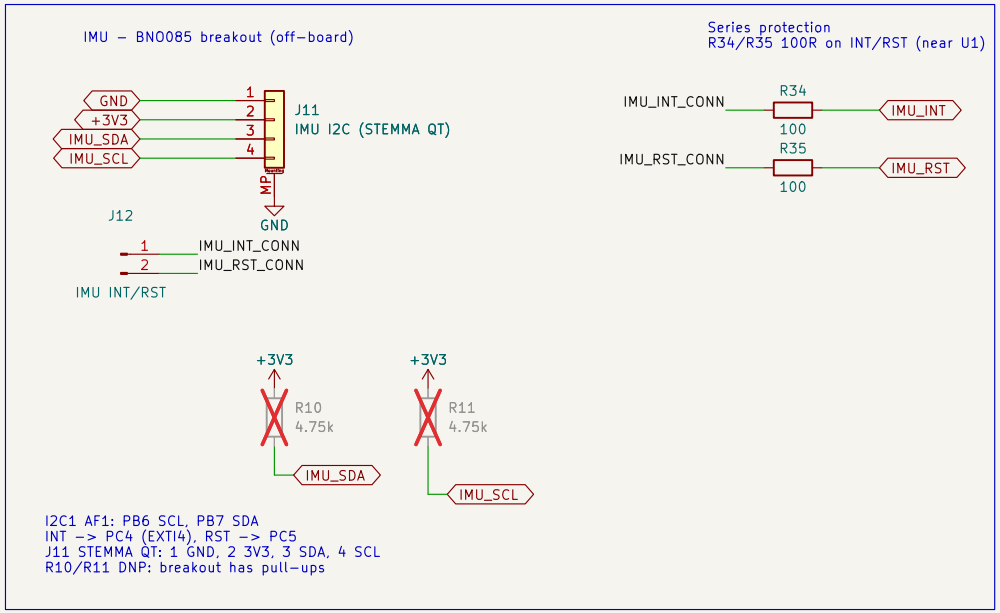




TP3: GND loop for the scope ground clip.
+5V_PI is on TP1 (Pi 5 Interface sheet),
used in the Pi bring-up check.

J5: one 2.54 mm header for every rail, ordered by voltage
1 GND
2 +3V3
3 +3.3VA (MCU analog)
4 +5V_SYS
5 +5V_AUX
6 +12V (after F1)

Senior Design		
Sheet: /Test Points/ File: Test_Points.kicad_sch		
Title: Power test points		
Size: A4	Date: 2026-10-08	**Rev: 1.1**
KiCad E.D.A. 10.0.6		Id: 10/11



Senior Design

Sheet: /IMU Interface/
File: IMU_Interface.kicad_sch

Title: IMU interface – BNO085 breakout (off-board)

Size: A4	Date: 2026-10-08	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 11/11